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Bengaluru - 560059, Karnataka, India

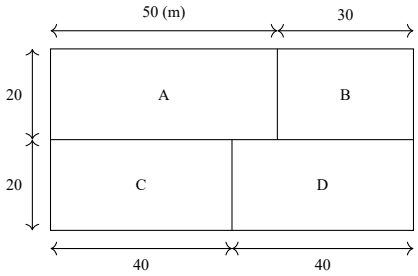
DEPARTMENT OF
# INDUSTRIAL ENGINEERING & MANAGEMENT

| | | |
|---|---|---|
| **Date : 25th May 2026** | **CIE – II** | **Max. Marks : 10 + 50** |
| **Semester : VI** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Facilities Planning And Design (Professional Core Elective)** | | **Course Code : IM365TDA** |

Note:
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | State the main objectives of material handling? | (02) | 1 | 1 |
| 2. | What is meant by unit load design? | (02) | 2 | 2 |
| 3. | Enumerate the importance of flexibility in material handling systems. | (02) | 3 | 3 |
| 4. | What is the role of conveyors in material handling? | (02) | 2 | 4 |
| 5. | State two objectives of computer aided layout techniques. | (02) | 2 | 3 |
| | **Part – B** | | | |
| 1. | Discuss the scope of material handling in modern industries. | (10) | 2 | 2 |
| 2. | Explain movement, time, quantity and space of a material handling system | (10) | 3 | 2 |
| 3. | Explain the following cost components used in estimating material handling cost:<br>i. Equipment cost<br>ii. Storage & Inventory Costs<br>iii. Damage/Wastage Costs | (10) | 3 | 3 |
| 4. | Explain the safety considerations to be followed in material handling operations. | (10) | 3 | 4 |

| | | | | |
|---|---|---|---|---|
| 5. | Consider the following data with unit cost matrix. Use CRAFT pair-wise interchange technique to obtain the desirable layout.<br>Total number of departments: 4<br>Total number of interchangeable departments: 4<br>Initial Layout: | (10) | 2 | 4 |




Area of Departments (Scale 1000 sq units = 1 sq unit):

| Department | A | B | C | D |
|---|---|---|---|---|
| Area (sq. units) | 10 | 6 | 8 | 8 |

Flow Matrix:

| From\To | A | B | C | D |
|---|---|---|---|---|
| A | – | 2 | 4 | 4 |
| B | 1 | – | 1 | 3 |
| C | 2 | 1 | – | 2 |
| D | 4 | 1 | 0 | – |

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DEPARTMENT OF
# INDUSTRIAL ENGINEERING & MANAGEMENT

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|---|---|---|
| **Date : 25-05-2026** | **CIE – II** | **Max. Marks : 10 + 50** |
| **Semester : VI** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Service Operations Management** | | **Course Code : IM365TDB** |

Note:
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | Imagine you are going to watch an IPL cricket match at Namma Bengaluru's very own Chinnaswamy Cricket stadium, list 4 aspects that you associate regarding service experience. | 2 | 1 | 1 |
| 2. | Assume you get in to an "IT enabled service" (ITES) industry in the next five years. Comment on how you intend to use Service as a competitive advantage. | 2 | 4 | 3 |
| 3. | Assume you are being interviewed for your short but successful career in the field of "Service operations management". One of the questions from a young enthusiast is "How to develop a culture of service quality in an organization?" Comment any two aspects. | 2 | 4 | 2 |
| 4. | Some of you have already started preparing for campus placements. In a year from now, many of your juniors would want your suggestions **(If placed)** and help in many matters of campus placements. To cater to this need and based on some of the requirement analysis that you have done, you wish to start a service and name it as "**Placement war room**" for engineering graduate students. Now answer the following questions.<br>1. Will it be online or offline and why?<br>2. What are the unmet needs of graduate students that your service will fulfil? | 2 | 4 | 4 |
| 5. | If your "**Placement war room**" idea clicks and is appreciated by your initial target audience. Identify any 2-success metrics of your service? Also, comment on the possible perception expectation gap for your service. | 2 | 4 | 3 |
| | **Part – B** | | | |
| 1 | Discuss the role of technology in developing the customer Experience. Cite two examples of services where technology has been used effectively to enhance customer experience. | 10 | 3 | 4 |
| 2 | The nature of supply chains can be very messy and complex. Describe the key issues service operations managers should focus on with a key focus on managing through intermediaries. | 10 | 3 | 4 |
| 3 | Write short notes on<br>a) Service level agreements (SLA)<br>b) Difference between SLA and Operational Level Agreements (OLA) | 5*2=10 | 2 | 3 |
| 4 | Envisage/visualize an experience map for a customer visiting a typical dental clinic. | 10 | 4 | 2 |

| | | | | |
|---|---|---|---|---|
| 5 | One of the key decisions for procurement function is choosing/selecting a supply partner and particularly for those who handle critical elements of supply chain. Discuss the criteria for supplier selection excluding Cost and Quality. | 10 | 3 | 4 |

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|---|---|---|
| **Date : 25/05/2026** | **CIE – II** | **Max. Marks : 10 + 50** |
| **Semester : VI** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Design of Experiments** | | **Course Code : IM365TDD** |

Instructions to Students:
1. Answer all questions
2. Use of statistical tables permitted

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | Define "Sparsity of Effects Principle." | 2 | 1 | CO1 |
| 2. | If you have a $2^5$ design, how many total treatment combinations are required for a single replicate? | 1 | 1 | CO1 |
| 3. | Compare the $2^2$ design and $3^2$ designs in terms of modeling. | 2 | 1 | CO2 |
| 4. | What is "Confounding" in factorial experiments? | 2 | 2 | CO2 |
| 5. | Define "Aliasing" in fractional factorial designs. | 1 | 2 | CO2 |
| 6. | Evaluate the risk of using a Resolution III design for a process where two-factor interactions are known to be significant. | 2 | 2 | CO2 |
| | **Part – B** | | | |
| 1 | An engineer is interested in the effects of cutting speed ($A$), tool geometry ($B$), and cutting angle ($C$) on the life (in hours) of a machine tool. Two levels of each factor are chosen, and three replicates of a $2^3$ factorial design are run. The results are as follows: | 20 | 2 | CO4 |

| $A$ | $B$ | $C$ | Treatment Combination | Replicates | | |
|---|---|---|---|---|---|---|
| | | | | I | II | III |
| − | − | − | (1) | 22 | 31 | 25 |
| + | − | − | a | 32 | 43 | 29 |
| − | + | − | b | 35 | 34 | 50 |
| + | + | − | ab | 55 | 47 | 46 |
| − | − | + | c | 44 | 45 | 38 |
| + | − | + | ac | 40 | 37 | 36 |
| − | + | + | bc | 60 | 50 | 54 |
| + | + | + | abc | 39 | 41 | 47 |

i. Estimate the factor effects. Which effects appear to be large?
ii. Use the analysis of variance to confirm your conclusions for part (i).

<table>
<tr><td>2</td><td>The effects of developer strength ($A$) and development time ($B$) on the density of photographic plate film are being studied. Three strengths and three times are used, and four replicates of a $3^2$ factorial experiment are run. The data from this experiment follow. Analyze the data using the standard methods for factorial experiments.
<table>
<tr><td colspan="2" rowspan="2"></td><td colspan="12">Development Time (minutes)</td></tr>
<tr><td colspan="4">10</td><td colspan="4">14</td><td colspan="4">18</td></tr>
<tr><td rowspan="3">Developer Strength</td><td>1</td><td>0</td><td>2</td><td>1</td><td>3</td><td>2</td><td>5</td><td>4</td><td>4</td><td>2</td><td>3</td><td>2</td><td>5</td></tr>
<tr><td>2</td><td>4</td><td>6</td><td>6</td><td>8</td><td>9</td><td>10</td><td>5</td><td>4</td><td>2</td><td>4</td><td>6</td><td>7</td></tr>
<tr><td>3</td><td>7</td><td>7</td><td>5</td><td>7</td><td>7</td><td>8</td><td>5</td><td>7</td><td>10</td><td>8</td><td>9</td><td>12</td></tr>
</table></td><td>20</td><td>3</td><td>CO4</td></tr>
<tr><td>3</td><td>List and explain the three key ideas / principles that are to be implemented for the successful use of fractional factorial designs.</td><td>10</td><td>2</td><td>CO2</td></tr>
</table>

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DEPARTMENT OF
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| | | |
|---|---|---|
| **Date : 6th June 2025** | **CIE – 2** | **Max. Marks : 10 + 50** |
| **Semester : VI** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: FACILITIES PLANNING AND DESIGN** | | **Course Code : IM365TDA** |

Note: Answer all the Questions.

**PART A**

| Q No.1 | Questions | M | BT | CO |
|---|---|---|---|---|
| 1.1 | Given station times:<br>• Station 1: 5 min<br>• Station 2: 4 min<br>• Station 3: 6 min<br>Find the bottleneck. | 01 | 2 | 1 |
| 1.2 | Total task time = 20 minutes, Number of workstations = 4, Cycle time = 6 minutes<br>Find line efficiency. | 02 | 1 | 1 |
| 1.3 | Calculate Throughput Time. Given:<br>• Station A: 2 min<br>• Station B: 3 min<br>• Station C: 4 min<br>What is the total throughput time for one product? | 02 | 2 | 1 |
| 1.4 | If a production unit has limited floor space and needs to allow a single operator to handle multiple machines easily, which layout (S-flow or U-flow) is better? | 01 | 1 | 1 |
| 1.5 | Define a unit load as per Tanchoco et al. | 02 | 2 | 1 |
| 1.6 | List any two basic methods of moving the unit load according to Apple. | 01 | 1 | 1 |

**Part –B**

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| 1 | State and explain the different types of material flow with suitable examples. | 10 | 2 | 1 |

<table>
<tr><td>2</td><td>A factory is designing a new assembly line for producing electric scooters. The production involves 5 sequential tasks with the following task times (in minutes):<br><table><tr><th>Task</th><th>Time (min)</th></tr><tr><td>A</td><td>4</td></tr><tr><td>B</td><td>6</td></tr><tr><td>C</td><td>5</td></tr><tr><td>D</td><td>3</td></tr><tr><td>E</td><td>2</td></tr></table>The management is considering either a Straight Line Flow layout or a U-Flow layout. Tasks:<br>a) Calculate the total time required to complete one unit.<br>b) Identify the bottleneck in the process.<br>c) If the total cycle time is 6 minutes and there are 3 workstations, compute the line efficiency.<br>d) Which layout (S-Flow or U-Flow) would you recommend if space is limited and one operator needs to handle multiple stations? Justify your answer</td><td>10</td><td>3</td><td>3</td></tr>
<tr><td>3</td><td>What are the Factors for Consideration in Planning Material Flow</td><td>10</td><td>2</td><td>2</td></tr>
<tr><td>4</td><td>Enumerate and elaborate the steps in material handling system design.</td><td>10</td><td>1</td><td>2</td></tr>
<tr><td>5</td><td>Explain the guidelines for estimating the material handling costs in practice.</td><td>10</td><td>1</td><td>1</td></tr>
</table>

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|---|---|---|
| **Date : 6-6-2025** | **CIE – 2** | **Max. Marks : 10 + 50** |
| **Semester : VI** | **UG** | **Duration : 30 + 90 Min** |
| **Course Title: Service Operations Management** | | **Course Code : IM365TDB** |

Note:
1. Answer all the Questions.

| Sl. No | Questions | M | BT | CO |
|---|---|---|---|---|
| | **Part – A** | | | |
| 1. | Think of a time when you recently received poor service. If service operations are so important, identify reasons why poor service is delivered sometimes? | 2 | 2 | 1 |
| 2. | Assume you get in to an "IT enabled service" (ITES) industry in the next five years. Comment on how you intend to use Service as a competitive advantage. | 2 | 4 | 3 |
| 3. | What is the service concept of "Cyber nurseries". | 2 | 2 | 2 |
| 4. | Comment on the importance of service process design. | 2 | 3 | 4 |
| 5. | From your experience, give two examples where technology is used to enhance customer experience. | 2 | 2 | 2 |
| | **Part – B** | | | |
| 1 | Discuss the role of technology in developing the customer Experience. Cite two examples of services where technology has been used effectively to enhance customer experience. | 10 | 3 | 4 |
| 2 | The nature of supply chains can be very messy and complex. Describe the key issues service operations managers should focus on with a key focus on managing through intermediaries. | 10 | 3 | 4 |
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| 4 | Visualize an experience map for a customer visiting a typical dental clinic. | 10 | 4 | 2 |
| 5 | One of the key decisions for procurement function is choosing/selecting a supply partner and particularly for those who handle critical elements of supply chain. Discuss the criteria for supplier selection excluding Cost and Quality. | 10 | 4 | 4 |